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CT Angiography with Spiral CT and Maximum Intensity Projection¹

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The authors describe a technique for obtaining angiographic images by means of spiral computed tomography (CT), preprocessing of reconstructed three-dimensional sections to suppress bone, and maximum intensity projection. The technique has some limitations, but preliminary results in 48 patients have shown excellent anatomic correlation with conventional angiography in studies of the abdomen, the circle of Willis in the brain, and the extracranial carotid arteries. With continued development and evaluation, CT angiography may prove useful as a screening tool or replacement for conventional angiography in some patients.

Index terms: Computed tomography (CT), spiral technology • Computed tomography (CT), three-dimensional, 1723.12987, 905.12987, 95.12987

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SPIRAL computed tomography (CT) permits acquisition of a large volume of data in seconds. It accomplishes this through collection of data with a continuously rotating gantry as the patient is passed through the scanning circle (1,2). From these data, multiple contiguous or overlapping sections of arbitrary thickness can be reconstructed. For example, data needed to reconstruct 30 contiguous 3-mm- (nominal) thick sections can be acquired through the abdomen in 30 seconds, resulting in 9 cm of coverage in a single breath hold. This acquisition speed permits capture of the arterial phase of a properly timed bolus of intravenous contrast material in

a large volume of patient anatomy. The inherent contrast available with CT, along with the possibility for reconstruction of cross sections without the superimposition of overlying structures, results in the possibility to perform CT angiography (CTA) with excellent visualization of vessel lumina, stenoses, and lesions, including aneurysms, atheromatous plaque, calcifications, and thromboses. We describe a technique for obtaining CTA data by using a spiral CT scanner and present the use of maximum intensity projection (MIP) for creating angiographic depictions of the carotid arteries and the arteries of the Circle of Willis and the abdomen.

Materials and Methods

To date, we have evaluated 48 patients with CTA. A Siemens Somatom Plus-S scanner was used in the spiral scanning mode in all cases (Siemens Medical Systems, Iselin, NJ). Nonionic contrast material (300 mg of iodine per milliliter) (Isovue; Squibb Diagnostics, New Brunswick, NJ) was injected into an antecubital fossa vein. For abdominal studies ($n = 22$), patients were instructed to hyperventilate prior to scanning and suspended ventilation for the duration of the acquisition. For the circle of Willis and carotid artery studies ($n = 26$), patients were instructed to breathe quietly during scanning. X-ray tube potential and current were set at 120 kVp and 165 mA, respectively. Continuous data were acquired for 30 seconds by using the table speeds, injection delays, injection rates, and spacing of reconstructed sections summarized in the Table. All images were reconstructed by using the standard reconstruction algorithm for the anatomic part being scanned.

Reconstructed images were transferred to an offline workstation (Sparc-2; Sun Microsystems, Mountain View, Calif) for preprocessing and MIP. MIP reconstructions in the superoinferior direction were performed for each set of reconstructed sections. Rectangular solid regions of interest (ROIs) for MIP reconstructions perpendicular to this direction were selected to suppress as much bone as possible and still include the vessels of interest. Separate regions were selected for the right and left carotid arteries. For the abdominal and circle of Willis studies, however, rectangular solid ROIs cannot be chosen to maximally eliminate bone without also removing important vessels. For

this reason, a semiautomated preprocessing algorithm was used to set all voxels in selected high-attenuation structures (eg, skull, spine) to zero. The algorithm identifies these structures on the bases of attenuation and spatial connectivity to user-selected "seed points" (3). This technique was also used to suppress calcific deposits from the walls of carotid arteries, to allow the MIP reconstructions to display the contrast medium column within the vessel lumen more clearly. MIPs perpendicular to the superoinferior axis and covering 180° of rotation in 6° increments were reconstructed from the preprocessed volumes and were displayed as cine loops. Preprocessing required 15 minutes of operator and 3 minutes of computer time per study, and MIP required approximately 10 seconds per projection angle.

Results

Figure 1 shows the results of a normal circle of Willis study obtained in a 43-year-old woman. The image is similar in appearance to a magnetic resonance (MR) image, which is the result of the suppression of data from the skull and bones of the base of the skull by means of the preprocessing algorithm. Arterial anatomy well beyond the A1 and M1 segments of the anterior and middle cerebral arteries could be seen with this technique. Preferential filling of the posterior dural sinuses (the transverse and sigmoid sinuses) can be seen on the axial collapsed image (Fig 1a) and the lateral MIP (Fig 1b). Contrast medium was not observed in the venous portion of the cavernous sinus with the injection protocol outlined in the Table. This allowed for clear definition of the margins of the carotid arteries in their intracavernous segments. ROIs may be selected to eliminate contrast medium-enhanced dural sinuses (as in the anteroposterior view shown in Fig 1c) or to selectively isolate right or left hemispheric or anterior or posterior features. Notice that incomplete suppression of bone edges resulted in contour artifacts. These artifacts, however, were easily differentiated from normal vascular anatomy.

Figure 2 shows the results of a carotid artery study performed in a 72-year-old man with an angiographically proved stenosis of 80% in the left carotid artery. The MIP clearly depicts the common, internal, and multiple branches of the external carotid artery. Jugular venous anatomy could also be clearly visualized

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Acquisition Parameters

Anatomic Feature	Table Speed (mm/sec)	Coverage (mm)	Injection Delay (sec)	Injection Rate (mL/sec)	Injection Duration (sec)	Section Thickness (mm)	Center-to-Center Spacing (mm)	No. of Sections
Abdomen	3	90	17-23	3-5	30	3	2	45
Circle of Willis	1	30	18	3	20	1	1	30
Carotid arteries	3	90	15	3	20	3	1	90

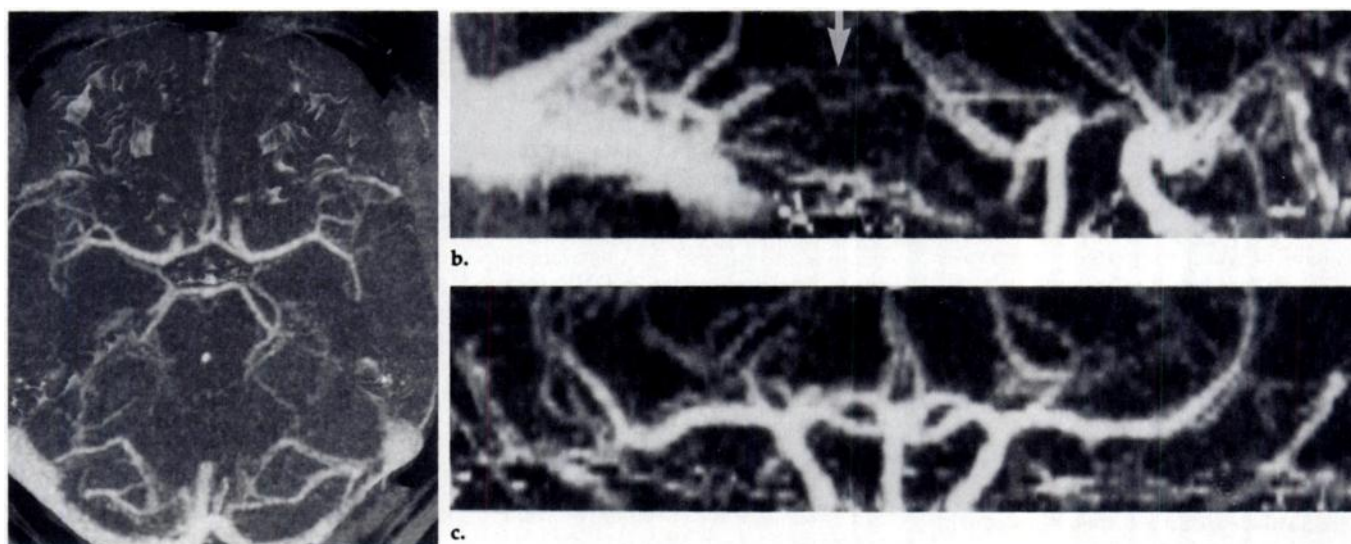


Figure 1. (a) Superoinferior, (b) lateral, and (c) anteroposterior MIPs through a 3-cm-thick slab encompassing the circle of Willis in a 43-year-old woman. The anterior "contour" artifacts in **a** were caused by incomplete removal of petrous bone edges. The anteroposterior projection (**c**) is through an ROI excluding anatomic features posterior to the arrow in **b**.

on these studies. Calcified regions within the carotid bulb are also shown; these can be differentiated from the contrast medium column on the basis of attenuation. Preprocessing allowed for suppression of carotid plaque calcifications in a manner analogous to suppression of other bone attenuation data, resulting in an image that reflected the contrast medium column alone (Fig 2c).

Figure 3a shows an anteroposterior MIP of the abdomen of a 76-year-old man, from CT data obtained with a 30-second breath hold and contrast medium injection rate of 3 mL/second. As with the carotid artery data, we preprocessed the data to suppress the spine. Several ribs have been left intact for reference, but these, too, could easily have been suppressed prior to MIP. The image clearly depicts the descending aorta, the superior mesenteric artery, the splenic artery, and the kidneys and renal arteries. Perfusion of the renal cortex obscures the intrarenal anatomy. Bilateral renal artery stenosis, however, is clearly evident (with associated calcification on the right side), as are aortic, superior mesenteric artery, and splenic artery calcifications. The spiral scanning sequence was performed in the superoinferior direction, and, as a result of suboptimal bolus timing and/or duration, contrast material was absent from the

aorta for the last 15% (4.5 seconds) of scanning. Findings at conventional angiography corroborated these results (Fig 3b).

Discussion

Display of the acquired images as conventional transaxial sections reveals the maximum detail inherent in the data. Anatomic features are difficult to discern in three dimensions with this format, however, and direct comparisons with conventional angiography are not possible. A wide variety of three-dimensional (3D) display techniques that result in images more directly comparable to angiograms are available (4-6). These methods include shaded surface display (7,8) and volume rendering (9-11). Shaded surface displays are generated by computing a mathematical model of a surface that connects all pixels with Hounsfield units greater than a predetermined threshold. The threshold must be carefully picked on the basis of contrast material attenuation in the area of interest, and in many cases, these displays result in clear depiction of vessel morphology. Because of suboptimal bolus timing, reduced flow distal to stenoses, and noise and partial volume effects, however, the attenuation in vessels may not be uniform. For

this reason, a volume-rendering technique, which does not reject any data because of a relationship to an arbitrary threshold, may be more suitable.

MIP is a volume-rendering method that is widely used for creation of MR angiography displays (12,13). With this technique, a two-dimensional image is built up through casting of mathematical rays in some desired viewing direction through a stack of reconstructed sections. The intensity of each pixel in the resulting image is the maximum intensity encountered along the ray as it traverses the volume. MIPs are computed from many view angles and can be displayed in a cine loop to convey 3D structure. MIP works well for MR angiography, because the pulse sequences used for data collection are designed to result in blood vessels having high intensities relative to those of nonvascular structures. In CTA, blood vessel attenuation is maximized by injection of contrast medium, but physical constraints preclude obtaining attenuation higher than that of bone and other calcified structures. The higher attenuation and spatial connectivity of bone, however, facilitates its selective removal from the data by means of preprocessing. The resulting high-contrast displays are directly comparable to conventional and MR angiograms. Additionally, the CTA

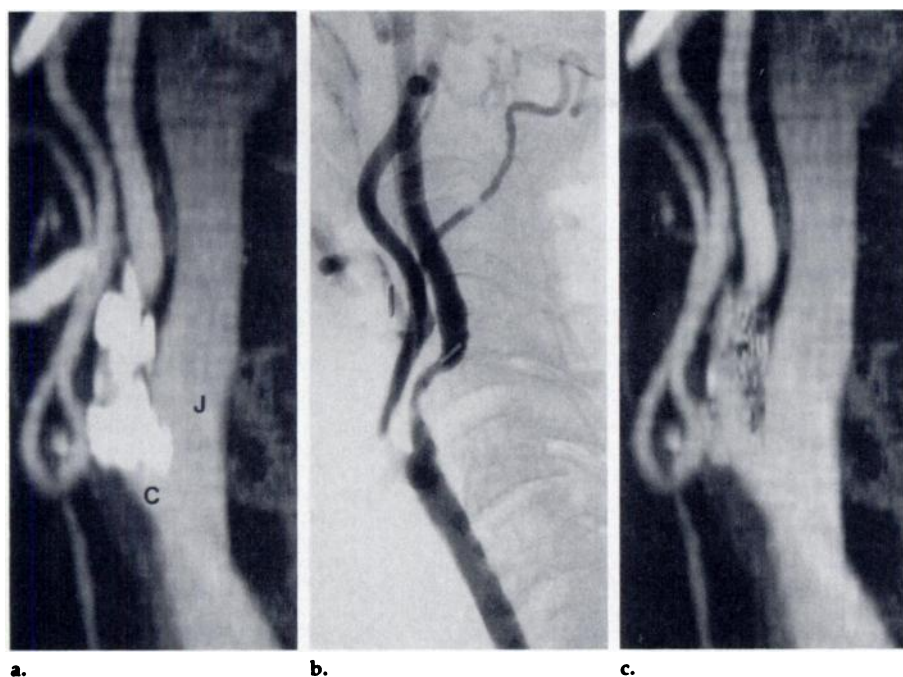
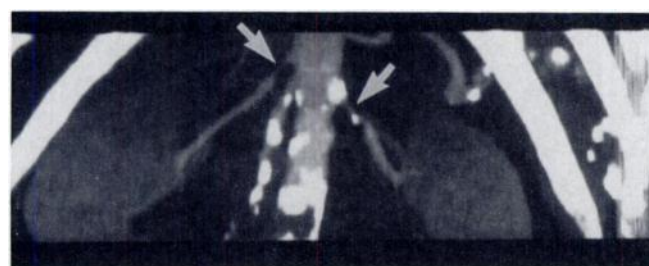
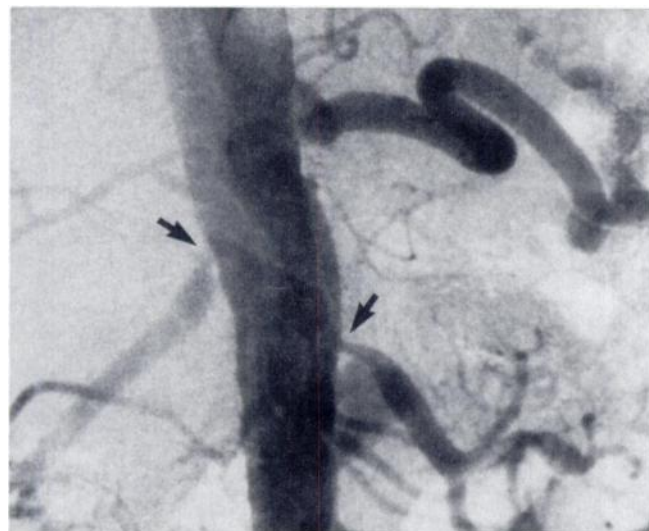


Figure 2. (a, c) MIPs and (b) angiogram obtained in a 72-year-old man with 80% stenosis of the left carotid artery. (a) Lateral MIP through a 9-cm-thick ROI encompassing the left carotid artery. The jugular vein (J) overlies the common carotid artery (C) proximal to the calcified carotid bulb. (b) An angiogram at an orientation similar to that in a shows 80% stenosis in the carotid bulb. (c) In this MIP, calcifications evident in a have been suppressed by preprocessing.



a.



b.

Figure 3. (a) Anteroposterior MIP through 9-cm-thick slab of abdomen and (b) angiogram obtained in a 76-year-old man. Renal stenosis (arrows) is evident. (a) Notice the calcifications in the right renal artery, aorta, superior mesenteric artery, and splenic artery.

displays can also be used to reveal the location and morphology of calcified plaques.

Because MIP projects the brightest voxels along each ray passing through

the volume, it is of prime importance to suppress all highly attenuating structures that are not of interest, such as the skull and spine. Direct methods such as slice editing (eg, wherein an operator

draws ROIs around structures to suppress) are effective but tedious and may not be reproducible. For small vessels, one pixel may represent a large fraction of the pixels in a luminal cross section, and a tracing error of even this magnitude may result in a false stenotic appearance.

We therefore desire a more automated method, such as the connectivity algorithm (3) currently employed. This algorithm starts with an operator-selected seed point within the structure of interest and proceeds to identify all voxels that are above a chosen Hounsfield unit threshold and spatially connected to the seed point itself or through other voxels that satisfy the threshold and connectivity requirements. Because of one or several possible CT artifacts (eg, partial volume or beam hardening) that may cause gradual instead of steep attenuation transitions between adjacent structures, however, this algorithm may include other structures as part of the structure one desires to suppress. For example, in the petrous carotid region, the vessels may physically touch the skull, and a connectivity algorithm using too low a threshold may falsely include the iodine-filled vessels as part of the skull.

A similar potential exists where the descending aorta lies against a portion of the spine and in attempts to suppress a vessel wall calcification that contacts the vessel lumen. In these cases, one is forced to choose a threshold well above the attenuation of the contrast medium in the vessel and thereby risk leaving bone edges unsuppressed because they have depressed CT numbers due to one of the effects mentioned above. An example of this is the contour artifact in Figure 1a. To some extent, this can be mitigated by use of the morphologic dilation operator (14) to extend the identified structure by a fixed distance along each of the three orthogonal axes, but care must be taken to ensure that vessels are not intruded upon. Also, the choice of CT reconstruction kernel may trade off some degree of bone edge enhancement for an increase in noise (15). Optimization of this operation is currently under investigation.

Conventional angiography offers superior spatial resolution, but CTA offers

more rapid examination time and lower radiation dose. In addition, the 3D nature of the acquired data makes it amenable to postprocessing (eg, volume-editing techniques such as manual cropping and automated connectivity-based algorithms) and reprojection from any angle. For example, although the common carotid artery is obscured by the overlying jugular vein in the projection shown in Figure 2, reprojection in the anteroposterior direction obviates this effect. In cases where an appropriate angle cannot be found, undesired structures can be removed by editing the volume prior to reprojection. With conventional angiography, on the other hand, predetermined views that may not reveal the desired anatomic relationships or vessel morphology are acquired. Additional contrast material and radiation are required to obtain additional views.

MR angiography may offer a noninvasive alternative to conventional angiography in some cases (12,13,16–21). Spatial resolution with MR angiography may be comparable to that with CTA, but acquisition times with MR angiography currently are too long to make breath holding (particularly desirable in the abdomen) possible. In addition, with MR angiography, vessel lumina are imaged indirectly by using velocities and/or tissue saturation effects, and complex flow patterns typically cause signal losses that can masquerade as stenoses or obscure pathologic entities. With CTA, the vessel lumina are delineated the same as with conventional angiography, that is, contrast material outlines the vessel lumen. One must be aware of CT artifacts such as beam hardening, scattered radiation, and partial volume, however, which can alter measured attenuation coefficients and decrease luminal opacity (22). These effects may be particularly important within calcified vessels and bony channels such as the cavernous carotid; this issue is a matter of current investigation.

Accomplishment of proper bolus timing in CTA is very important (15). Suboptimal timing may result in deficient or reduced vessel opacification and/or obscuration of the desired features by or-

gan parenchyma and/or veins. In some cases, it may be possible to target MIPs to tightly cropped ROIs and/or employ the same preprocessing algorithm used for bone and calcification suppression to suppress veins, but parenchymal enhancement obscuring intraorgan arteries may be more difficult to overcome.

In conclusion, we have demonstrated the potential of CTA performed with spiral scanning technology, MIP, and intravenous contrast enhancement in the circle of Willis, the carotid bifurcation, and the abdomen. Initial correlation to angiographically proved stenoses has been excellent. Calcifications present in vessel walls are clearly visualized or may be mathematically suppressed if desired. In comparison with conventional angiography, CTA is performed more rapidly, at less expense, and with reduced radiation dose, and ROIs and projection angles may be retrospectively chosen to demonstrate anatomic features to best advantage. While prospective studies comparing CTA, MR angiography, and conventional angiography are required, CTA has the potential to become a minimally invasive screening tool and obviate the need for conventional angiography in some patients. ■

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References

1. Kalender WA, Polacin A. Physical performance characteristics of spiral CT scanning. *Med Phys* 1991; 18:910–915.
2. Villafana T. Technologic advances in computed tomography. *Curr Opin Radiol* 1991; 3:275–283.
3. Cline HE, Dumoulin CL, Hart HR Jr, Lorensen WE, Ludke S. 3D reconstruction of the brain from magnetic resonance images using a connectivity algorithm. *Magn Reson Imaging* 1987; 5:345–352.
4. Mankovich NJ, Robertson DR, Cheeseman AM. Three-dimensional image display in medicine. *J Digit Imaging* 1990; 3:69–80.
5. Strong AB, Lobregt S, Zonneveld FW. Applications of three-dimensional display techniques in medical imaging. *J Biomed Eng* 1990; 12:233–238.
6. Henri CJ, Pike GB, Collins DL, Peters TM. Three-dimensional display of cortical anatomy and vasculature: magnetic resonance angiography versus multimodality integration. *J Digit Imaging* 1991; 4:21–27.

7. Magnusson M, Lenz R, Danielsson PE. Evaluation of methods for shaded surface display of CT volumes. *Comput Med Imaging Graph* 1991; 15:247–256.
8. Cline HE, Lorensen WE, Souza SP, et al. 3D surface rendered MR images of the brain and its vasculature. *J Comput Assist Tomogr* 1991; 15:344–351.
9. Davis RE, Levoy M, Rosenman JG, et al. Three-dimensional high-resolution volume rendering (HRVR) of computed tomography data: applications to otolaryngology-head and neck surgery. *Laryngoscope* 1991; 101:573–582.
10. Levoy M. Methods for improving the efficiency and versatility of volume rendering. *Prog Clin Biol Res* 1991; 363:473–488.
11. Fishman EK, Magid D, Ney DR, Drebin RA, Kuhlman JE. Three-dimensional imaging and display of musculoskeletal anatomy. *J Comput Assist Tomogr* 1988; 12:465–467.
12. Keller PJ, Drayer BP, Fram EK, Williams KD, Dumoulin CL, Souza SP. MR angiography with two-dimensional acquisition and three-dimensional display. *Radiology* 1989; 173:527–532.
13. Laub GA. MR angiography with gradient motion refocussing. *J Comput Assist Tomogr* 1988; 12:377–382.
14. Serra J. Image analysis and mathematical morphology. New York: Academic Press, 1982.
15. Rubin GD, Dake MD, Napel S, McDonnell CH, Jeffrey RB Jr. Abdominal 3D spiral CT angiography: preliminary experience. *Radiology* (in press).
16. Dumoulin CL, Hart HR. Magnetic resonance angiography. *Radiology* 1986; 161:717–720.
17. Dumoulin CL, Klein HE, Souza SP, Wagle W, Walker MF. Three-dimensional time-of-flight magnetic resonance angiography using spin saturation. *Magn Reson Med* 1989; 11:35–46.
18. Dumoulin CL, Souza SP, Walker MF, Wagle W. Three-dimensional phase-contrast angiography. *Magn Reson Med* 1989; 9:139–149.
19. Nishimura DG, Macovski A, Pauly J. MR angiography by selective inversion recovery. *Magn Reson Med* 1987; 4:193–202.
20. Parker DL, Yuan C, Blatter DD. MR angiography by multiple thin slab 3D acquisition. *Magn Reson Med* 1991; 17:434–451.
21. Ruggieri PM, Laub GA, Masaryk TJ, Modic MT. Intracranial circulation: pulse sequence considerations in three-dimensional (volume) MR angiography. *Radiology* 1989; 171:785–791.
22. Joseph P. Artifacts in computed tomography. In: Newton TH, Potts DG, eds. *Radiology of the skull and brain: technical aspects of computed tomography*. St. Louis: Mosby, 1981.